

MAVERICK FORENSIC INCIDENT REPORT

Case ID: MAV-20260919-064925-670F5315 | Generated: 2026-09-19T06:49:25.681065+00:00 | Platform: MAVERICK 1.0.0

1. Case Summary & Threat Assessment

VERDICT: MEDIUM	Risk Score: 0.5946 / 1.0000		
	Threat Indicators Flagged: 4		
	Summary: Overall threat assessed as MEDIUM (Risk Score: 0.59/1.00) with 4 suspicious/malicious indicators identified across forensics modules.		
Subject:	URGENT: Account Suspension Notice - Action Required		
From:	Account Security Team		
To:	user@corporate-target.com		
Date:	Sat, 19 Sep 2026 06:00:00 +0000		
Message-ID:			
Chain of Custody SHA-256:	670f53154dbe1c2e17872882f73703029854a83aa5257fd68ce05e949462e12a		

2. Evidence Fusion Scoring Breakdown

Forensic Dimension	Assigned Weight	Weighted Score	Contribution / Status
Machine Learning Phishing	40% (max 0.40)	0.3946	Lexical vector probability & term attribution
Email Authentication Forensics	20% (max 0.20)	0.1000	SPF / DKIM / DMARC status and From: domain alignment
Attachment Static Forensics	20% (max 0.20)	0.0000	MIME type magic bytes, VBA macros, and PE headers
IOC & Geolocation Signals	20% (max 0.20)	0.1000	Raw IP URLs, high-abuse TLDs, and routing anomalies

Key Contributing Factors:

- Machine Learning: High confidence phishing text detected (98.7% probability) (prominent terms: click, account, money, hours).
- Authentication: Sender identity misalignment detected — From: 'paypal-update-alert.com' does not align with authenticated SPF or DKIM domains.
- Attachments: No email attachments present.
- IOC Signals: Identified communication with high-abuse top-level domain(s): .xyz.

3. Machine Learning Phishing Classification Findings

Phishing Probability:	98.66%	Predicted Label:	PHISHING	Model Confidence:	98.7%
Influential Lexical Term	Indicator Class	Model Feature Weight	TF-IDF Weight	Directional Impact	
click	PHISHING	5.9536	0.2101	1.2510	
account	PHISHING	1.8196	0.3692	0.6718	
money	PHISHING	3.3856	0.1306	0.4422	
hours	PHISHING	1.8758	0.1578	0.2960	
immediately	PHISHING	1.4093	0.1679	0.2367	
permanent	PHISHING	0.9598	0.2245	0.2154	

4. Email Authentication Forensics & Domain Alignment

Protocol	Result Status	Identifier Alignment	Evaluated Domain(s)
SPF	NONE	MISALIGNED	[None]
DKIM	NONE	MISALIGNED	[None]
DMARC	NONE	FAIL	From: paypal-update-alert.com

Authentication Notes:

- No Authentication-Results headers present. Triggering direct SPF/DKIM verification.
- Direct DKIM verification: No 'DKIM-Signature' header present on email.
- Direct SPF verification: Could not extract client IP from Received hop chain.
- SPF alignment note: No SPF Return-Path / MailFrom domain found to compare.

5. Indicators of Compromise (IOCs) & Excluded Subnets

Category	Forensic IOC Values
Public IPs	[None]
Excluded IPs (RFC 1918)	[None]
Target Domains	paypal-security-verification.xyz
Extracted URLs	https://paypal-security-verification.xyz/verify
Mentioned Emails	[None]

6. Geolocation Intelligence Telemetry

No public IP geolocation metadata returned.

7. Attachment Static Forensics

No email attachments present in message.

8. Investigation Timeline (Received Hop Chain)

No Received headers parsed in MTA chain.

9. Incident Remediation & Forensic Recommendations

Priority	Category	Recommended Action & Evidentiary Rationale
HIGH	Identity & Authentication	Block sender identity and review DMARC enforcement for domain 'paypal-update-alert.com'. <i>Rationale:</i> DMARC verification failed or sender identity misalignment indicates active email spoofing / impersonation.
CRITICAL	Credential & User Action	Enforce mandatory password reset and revoke active SSO/OAuth session tokens for recipients. <i>Rationale:</i> ML phishing classifier detected high-confidence phishing language (98.7% probability) targeting user credentials.
MEDIUM	Network & Firewall	Add high-abuse domain(s) (paypal-security-verification.xyz) to DNS sinkhole / protective resolver blocklist. <i>Rationale:</i> Domain infrastructure is registered under known high-abuse / disposable TLDs.